

20. (1.25 points)

Given the following financial information (\$000) for an insurance company as of December 31, 2016:

Beginning US GAAP equity	100,000
Net income	20,000
Minimum capital to maintain S&P AA rating	110,000
Minimum capital to meet management's year-end growth target	115,000
Change in loss and expense reserves	25,000
Advertising expenses (cash charge)	3,000
Amortized office furnishings expense	2,000
Net borrowing	10,000

a. (0.75 point)

Determine the company's Free Cash Flow to Equity as of year-end.

b. (0.5 point)

Briefly describe two reasons why the Free Cash Flow to Equity method is preferred over the Free Cash Flow to the Firm method when valuing property and casualty insurance companies.

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 20	
TOTAL POINT VALUE: 1.25	LEARNING OBJECTIVE(S): B1, B2
SAMPLE ANSWERS	
Part a: 0.75 point	
<u>Sample 1</u> FCFE = net income + non-cash charge - net working capital - change in required capital + net borrowing Use the maximum of the two min-capital requirements, 115. $FCFE = 20,000 + 2,000 - (115,000 - 100,000) + 10,000$ $FCFE = 17,000$ <u>Sample 2</u> $20 + 2 - 15 + 10 = 17$	
Part b: 0.50 point	
<u>Sample 1</u> 1. Free cash flow to firm indirectly calculates the value by calculating total cash flow to the firm and subtracting debt. The difference between debt and policyholders liability is arbitrary. 2. Free cash flow to firm also requires calculating the weighted average cost of capital; this is complicated by the same issue as 1 -> the split between debt and PH liability is arbitrary but needed when calculating the WACC. <u>Sample 2</u> 1. FCFF calculates the equity value by considering FCF to the firm -> firm value and then subtract the market value of the debt to get equity value. Since insurance companies have policyholder liabilities it's hard to distinguish that from debt issued, we have trouble figuring out how to treat PHL when subtracting debt value. 2. FCFF requires either the weighted average cost of capital (reflects risk to both debtholders and equity holders) or the all equity discount rate to discount the CFs. Again, due to the difficulty in distinguishing debt vs PHL, it's hard to come up with / calculate an appropriate discount rate for FCFF. <u>Sample 3</u> Due to policyholder liabilities, hard to calculate debt as needed for FCFF, or to calculate weighted average cost of capital for discounting. <u>Sample 4</u> Free cash flow to firm (FCFF) uses the weighted average cost of capital (WACC) since cash flows come from both equity and debt capital. Since it's hard to differentiate policyholder liabilities from other sources of debt, it's hard to precisely define WACC. Free cash flow to equity doesn't rely on WACC so it's not an issue. FCFF also relies on adjusted present value (APV) which is also hard to define because it's hard to differentiate policyholder liabilities from other debt. FCFE doesn't need this.	

EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

Sample 5

Free cash flow to firm (FCFF) takes the total value of the firm then subtracts off debt. It is hard to distinguish policyholder liability from other debt.

FCFF uses a weighted average cost of capital which requires the average duration of the debt, which is hard to calculate for policyholder liabilities.

EXAMINER'S REPORT

Candidates were expected to understand the Free Cash Flow to Equity method for valuing the equity of a P&C insurer.

Part a

Candidates were expected to correctly identify and use the financial information provided to determine the Free Cash Flow to Equity as of year-end.

Common errors include:

- Incorrectly treating of amortized office furnishings expense, either by:
 - Excluding it completely
 - Correctly identifying it as a non-cash charge, but then incorrectly subtracting it instead of adding it
 - Incorrectly identifying it as a working capital expense/investment, thus subtracting it.
- Including the advertising expenses (cash charge) in any way.
- Failing to use the more binding (larger) capital constraint when determining the change in required capital.
- Dividing by anything to get a quotient as a final answer.

Part b

Candidates were expected to explain that the arbitrary distinction between traditional debt and a P&C insurer's policyholder liabilities (PHL) complicates the free cash flow to the firm (FCFF) method in two ways: (1) the discount rate used to value the firm in total and (2) the subtraction of the market value of debt.

Common errors include:

- Simply defining the FCFF and/or FCFE methods.
- Stating that the FCFE method is an easier/more direct calculation, without further explanation of what makes the FCFF method more challenging for a P&C insurer.
- Stating that the FCFE method is somehow a better representation of shareholder value, or more aligned with shareholders perspectives.
- Stating that the weighted average cost of capital (WACC) is "hard/harder/impossible to calculate" (or similar) without explaining why. Need to mention the arbitrary distinction between debt/PHL.
- Stating that the WACC is harder to calculate because it is a combined discount rate for equity and debt, without explaining why the addition of debt makes the calculation more challenging. Again, need to mention the PHL for a P&C insurer.
- Simply stating that the FCFF method "does not include policyholder liabilities" (or similar) without explaining that the FCFF method values the entire firm then subtracts the market value of debt, or without mentioning either the arbitrary distinction between PHL and

EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

debt, or a lack of economic rational to treat PHL differently from debt.

- Simply stating that the policyholder liabilities are “hard to value”, (or similar) without explaining that the FCFF method values the entire firm then subtracts the market value of debt, or without mentioning either the arbitrary distinction between PHL and debt, or a lack of economic rational to treat PHL differently from debt.
- Stating that either method uses measurements that management is more familiar with, or are more readily available from financial statements.
- Confusing policyholders liabilities with policyholders surplus or equity.